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Exercise 1: Critical Paper Review

Q1. The paper addresses a gap in NLP resources for detecting subtle, harmful language: prior to this work, no large-scale dataset existed for Patronising and Condescending Language (PCL) directed at vulnerable communities. The primary contributions are: (i) the *Don't Patronize Me!* dataset of 10,637 news paragraphs, annotated at two levels: paragraph-level binary PCL presence, and span-level annotation identifying the specific words or phrases responsible; (ii) a two-level taxonomy of PCL with 7 fine-grained categories (e.g., *The Saviour*, *Compassion*, *Authority voice*); and (iii) baseline experiments across a range of models, including transformer-based ones, making the dataset immediately usable as a benchmark. The dataset was subsequently released as the SemEval 2022 Task 4 shared task, enabling community-wide evaluation.

Q2. A key technical strength is the annotation pipeline. Given the inherent subjectivity of PCL, the authors use a principled two-stage process: paragraph-level binary labeling first, followed by span and category annotation only for confirmed positives. They report inter-annotator agreement at each stage and employ a referee annotator to adjudicate total disagreements. This directly mitigates label noise in a task where single-pass annotation would be unreliable.

A second strength is the immediate benchmarkability of the dataset. The authors define two concrete tasks (binary PCL detection and fine-grained category prediction), evaluate a range of models including transformer-based ones via cross-validation, and show that even strong baselines struggle (F1 ≈ 0.49 on binary detection). This not only gives future work a clear starting point, but also quantitatively demonstrates that the task poses a genuine and non-trivial research challenge.

Q3. The most significant weakness is the moderate inter-annotator agreement reported in several settings. For a subjective phenomenon like PCL, this is understandable, but it has a concrete consequence that the paper does not fully address: if annotator agreement is moderate, small differences in model F1 may fall within the noise floor of the labels themselves, making it difficult to draw confident conclusions about which approach is genuinely better. The paper does not discuss this limitation or provide confidence intervals around its baseline results.

A second weakness is that the span-level annotations are never fully exploited in the evaluation. Despite being a central contribution, the paper does not evaluate any method on the task of predicting exact spans and their categories — the baseline experiments treat the problem at paragraph level only. Additionally, several taxonomy categories are rare and yield poor baseline performance, but the paper largely observes this without offering

analysis of why or suggesting mitigation strategies such as data augmentation or targeted collection.

Exercise 2: Exploratory Data Analysis

EDA Technique 1: Class balance (train vs dev). Visual/Tabular Evidence: Table 1 reports class counts and the PCL positive rate for each split, and Figure 1 visualizes the same distribution.

Table 1: Class distribution and PCL positive rate for the official train/dev splits.

Split	n_{total}	$n_{\text{No PCL}}$	n_{PCL}	PCL rate
train	8375	7581	794	0.094806
dev	2094	1895	199	0.095033

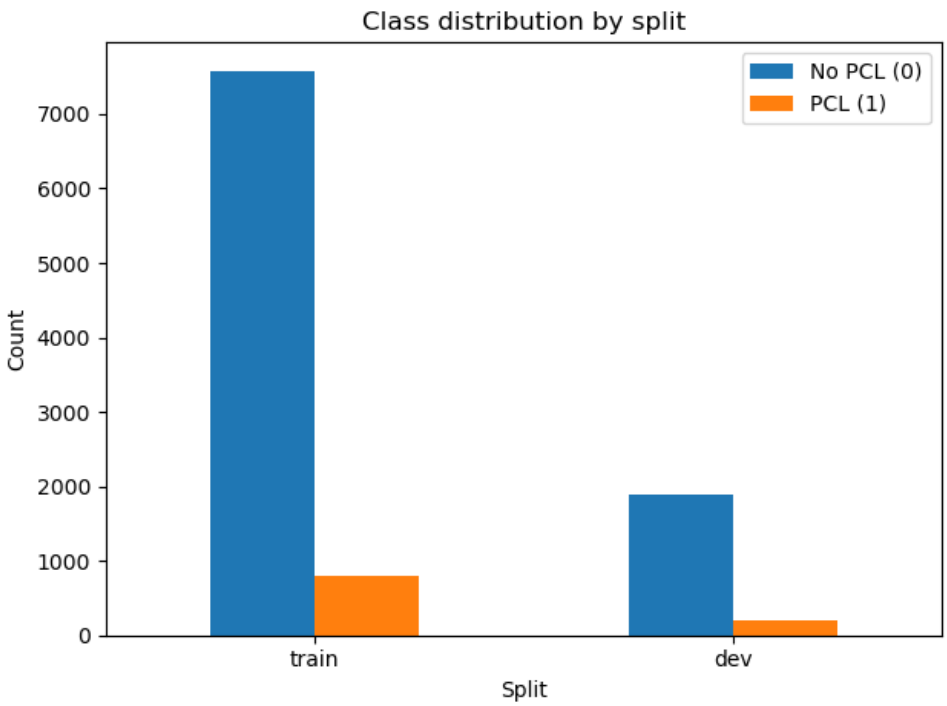


Figure 1: Class distribution by split (0 = No PCL, 1 = PCL).

Analysis: The dataset is clearly imbalanced: in train there are 7,581 No-PCL examples and 794 PCL examples (PCL rate ≈ 0.0948), while dev has 1,895 No-PCL and 199 PCL (PCL rate ≈ 0.0950). Only about 9.5% of the data is positive, but the positive rate is nearly identical across splits, which suggests the official train/dev partition is well-stratified.

Impact Statement: Because of this imbalance, we do not rely on accuracy as the main metric (a majority-class predictor would score $\approx 90.5\%$ accuracy trivially). Instead, we focus on positive-class F1, use stratified splits throughout, and apply class weighting and decision threshold tuning rather than defaulting to 0.5, since the base rate is low.

EDA Technique 2: Keyword/topic skew (PCL rate by keyword). Visual/Tabular Evidence: Table 2 summarizes PCL rates per keyword in the training

split, and Figure 2 shows the same values as a bar plot. Table 3 compares train vs dev keyword-level PCL rates and reports their absolute difference.

Table 2: PCL rate by keyword in the training split (all keywords shown, each with $n \geq 50$).

Keyword	n	PCL rate
homeless	865	0.172254
in-need	856	0.167056
poor-families	719	0.155772
hopeless	788	0.124365
refugee	880	0.082955
disabled	834	0.080336
vulnerable	871	0.068886
women	837	0.045400
migrant	882	0.035147
immigrant	843	0.027284

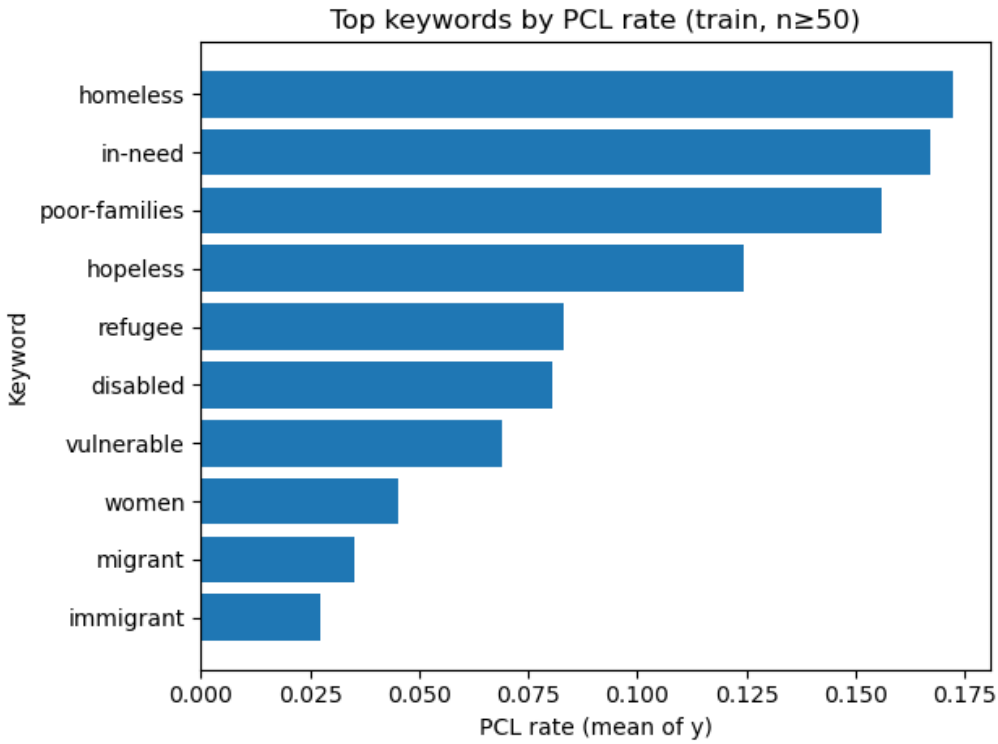


Figure 2: PCL rate (mean of y) by keyword in the training split.

Analysis: The probability of PCL varies substantially across vulnerability keywords (Table 2). For example, `homeless` and `in-need` have relatively high PCL rates (≈ 0.17), whereas `immigrant` and `migrant` are much lower (≈ 0.03). Crucially, these rates are highly stable between splits: the correlation of keyword-level PCL rates between train and dev is ≈ 0.91 (Table 3). This is not merely a data collection artefact — it likely reflects something genuine about how different vulnerable groups are discussed in news media, with some groups (e.g. the homeless, those in need) being more frequently written about in a patronizing register than others (e.g. immigrants, migrants).

Table 3: Train vs dev keyword-level PCL rate comparison (correlation ≈ 0.91).

Keyword	n_{train}	$\text{PCL}_{\text{train}}$	n_{dev}	PCL_{dev}	$\lvert\Delta\rvert$
poor-families	719	0.155772	190	0.200000	0.044228
homeless	865	0.172254	212	0.136792	0.035462
vulnerable	871	0.068886	209	0.095694	0.026807
in-need	856	0.167056	226	0.146018	0.021038
women	837	0.045400	233	0.060086	0.014686
refugee	880	0.082955	188	0.069149	0.013806
migrant	882	0.035147	207	0.024155	0.010993
disabled	834	0.080336	194	0.072165	0.008171
immigrant	843	0.027284	218	0.032110	0.004827
hopeless	788	0.124365	217	0.119816	0.004550

Impact Statement: This skew means a model could exploit keyword identity as a shortcut, learning that certain group labels are "more PCL" rather than genuinely modelling patronizing language. To guard against this, we track performance broken down by keyword throughout our error analysis, and use regularisation and robust validation during training. The keyword slice analysis later confirms this concern: `in-need` achieves F1 = 0.79 while `women` and `immigrant` reach only 0.36 and 0.44 respectively, suggesting the model has indeed learned keyword-correlated shortcuts for some groups.

Exercise 3: Baseline model and proposed novel approach

Proposed approach (BestModel). Rather than swapping in a bigger backbone, we kept RoBERTa-base and focused on *how* we train and evaluate it. The core additions:

Pipeline (BestModel)
Input: paragraph text → RoBERTa-base encoder → shared representation
→ **Binary head** (2-class logits)
→ **Taxonomy head** (7 sigmoid logits; multi-label)
Training loss: $\mathcal{L} = \mathcal{L}_{bin} + \lambda\,\mathcal{L}_{tax}$ (taxonomy loss on PCL-positive examples only)
Inference: $\hat{y} = \mathbb{1}\{p(\text{PCL}) \geq \tau\}$ where τ is tuned on the dev set.

Figure 3: Overview of the proposed approach.

1. **Multi-task taxonomy supervision.** The dataset labels each PCL example with a 7-category taxonomy (e.g., "Saviour", "Unequal power framing"). We add an auxiliary head that predicts these categories jointly with the binary label — computed only on positive examples to avoid the all-zero negatives drowning out the signal. The intuition is that learning *how* PCL manifests should sharpen the binary decision boundary.

2. **Positive-class weighting and threshold calibration.** We up-weight the loss on PCL examples (w_+) to give stronger gradient signal from the scarce positives. Combined with sweeping the decision threshold τ on the dev set to directly maximise F1, this lets us trade precision for recall in a controlled way. With only ∼9.5% positives, the default 0.5 threshold is a poor fit for maximising F1.

3. **Linear LR schedule with warmup.** We use a linear decay schedule with a 6% warmup ratio rather than a fixed learning rate, which stabilises early training and avoids large gradient updates at the start of fine-tuning.
4. **Hyperparameter selection via 5-fold stratified CV.** All hyperparameter decisions — learning rate, `pos_weight`, `category_loss_weight`, weight decay, and number of epochs — were made using stratified k-fold cross-validation on the training set only. The official dev set was held out entirely until final evaluation, ensuring no information leakage from hyperparameter selection.
5. **Train+dev retraining for test predictions.** For the final test submission, we retrained on the combined train and dev sets using the best hyperparameters found in CV, giving the test model maximum signal without compromising the integrity of the dev evaluation.

Rationale and expected outcome. PCL is subtle, the data is imbalanced, and positive-class F1 is sensitive to the decision threshold. Each component directly targets one of these problems: richer representations from multi-task learning, better recall from class weighting and threshold tuning, stable convergence from the LR schedule, and principled hyperparameter choices from CV.

Exercise 5: Local Evaluation and Error Analysis

We conduct local evaluation on the official development set (2093 examples; 199 positives and 1894 negatives) to understand model behaviour beyond the leaderboard score. Our submitted model is a RoBERTa-base multi-task classifier trained with auxiliary taxonomy supervision and imbalance-aware optimisation (positive weighting), evaluated using a calibrated decision threshold.

Overall performance on the dev set

Following the shared-task protocol, we report **positive-class F1** (PCL = 1) on the official dev set. We tune the decision threshold τ on the dev set to maximise positive-class F1. The best threshold found was $\tau = 0.10$, yielding the results in Table 4.

Table 4: Global dev-set performance at tuned threshold $\tau = 0.10$.

Threshold τ	Precision (PCL)	Recall (PCL)	F1 (PCL)	Accuracy
0.10	0.6368	0.6080	0.6221	0.9298

Table 5 shows the confusion matrix at $\tau = 0.10$. The model correctly identifies 121 of 199 PCL examples while producing 69 false alarms on the dominant negative class.

Table 5: Confusion matrix on dev at $\tau = 0.10$.

	Pred 0 (No PCL)	Pred 1 (PCL)
True 0 (No PCL)	1825	69
True 1 (PCL)	78	121

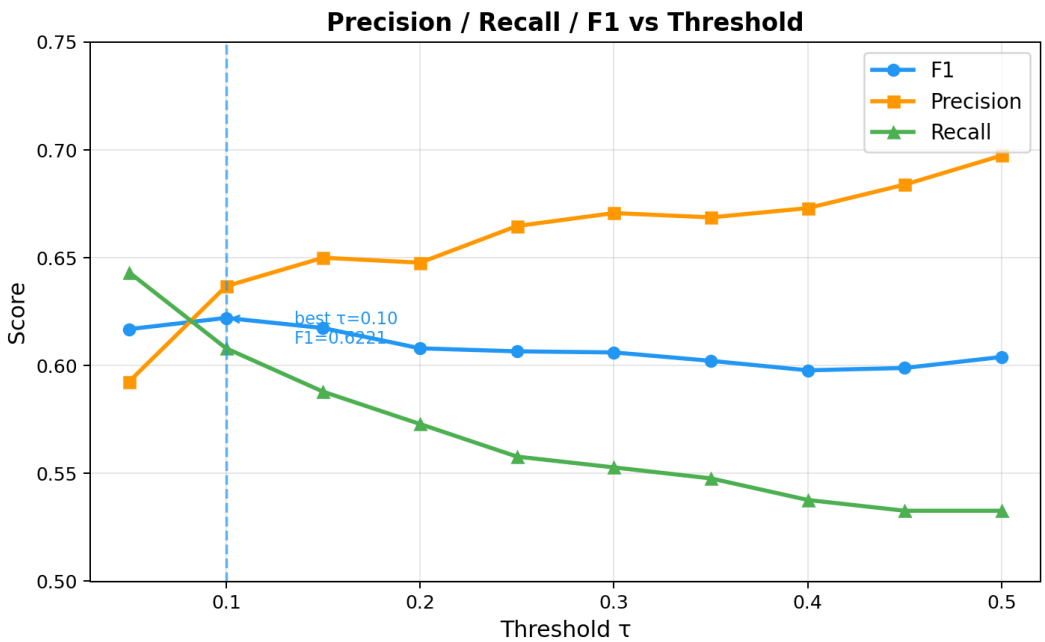
Interpretation. The tuned threshold $\tau = 0.10$ is substantially lower than the default 0.5, reflecting both class imbalance and the fact that F1-optimal thresholds shift downward when recall must be emphasised. With only ∼9.5% positives, a lower threshold is necessary to achieve a favourable precision–recall trade-off for the positive class.

Threshold sensitivity analysis

Table 6 reports the top-performing thresholds from a sweep over $[0.05, 0.50]$. Performance is relatively stable across a broad neighbourhood ($\tau \in [0.05, 0.30]$), with F1 declining only gradually as τ increases. This indicates the model is well-calibrated and not brittle to the exact threshold choice.

Table 6: Top thresholds by dev F1. Best threshold is $\tau = 0.10$.

τ	F1 (PCL)	Precision (PCL)	Recall (PCL)	% predicted positive
0.10	0.6221	0.6368	0.6080	9.08%
0.15	0.6174	0.6500	0.5879	8.60%
0.05	0.6169	0.5926	0.6432	10.32%
0.20	0.6080	0.6477	0.5729	8.41%
0.25	0.6066	0.6647	0.5578	7.98%
0.30	0.6061	0.6707	0.5528	7.84%
0.50	0.6040	0.6974	0.5327	7.26%



Qualitative error analysis

We analyse **high-confidence** false positives (FP) and false negatives (FN) — cases where $p(\text{PCL})$ is extreme but the prediction is wrong. These reveal systematic confusions rather than borderline threshold effects.

High-confidence false positives (FP). A recurring FP pattern is that the model over-triggers on *vulnerability/aid* lexical cues (e.g., `in-need`, `hopeless`, `refugee`, `homeless`). Many of these texts describe support, charity, or religious sentiment in a neutral or inspirational tone, which superficially resembles PCL-associated language without actually being patronizing. This is consistent with the low τ and positive weighting setup, which intentionally favours recall and can increase false alarms on "helping" contexts.

High-confidence false negatives (FN). High-confidence FNs often contain PCL cues that are *indirect* or embedded in factual or policy-reporting language. These instances lack overt "helping" lexicon, or express patronization through subtle framing and discourse-level implications, leading the model to assign very low $p(\text{PCL})$ despite the positive label.

Representative high-confidence errors. Tables 7 and 8 show the five highest-confidence errors of each type. Full texts are provided in the Appendix.

Table 7: Representative high-confidence false positives. Bold indicates the likely trigger cue.

ID	Keyword	Ctry	$p(\text{PCL})$	Snippet
FP1	in-need	ph	0.9926	...a gift, being a nurse ... **to help those who are in need** ...
FP2	hopeless	ph	0.9899	...share her skills to build on the dreams of women and children ... **especially to those who are hopeless**
FP3	in-need	my	0.9893	...Jesus begins his teaching ... **those in need of comfort** ... blessed are they who mourn
FP4	in-need	in	0.9888	...show unconditional love ... **helping those who are in need of our help and care**
FP5	refugee	ke	0.9869	...a clinic called "Hope" **helps a Syrian refugee boy** cope with diabetes

Table 8: Representative high-confidence false negatives. Bold indicates the likely missed cue.

ID	Keyword	Ctry	p(PCL)	Snippet
FN1	vulnerable	ng	0.0010	...surcharge to **protect the poor and vulnerable amongst us** ...
FN2	disabled	hk	0.0010	... **20 disabled undergraduate students** ...receiving the same salaries as able-bodied colleagues
FN3	poor-families	us	0.0011	...disease hit areas where **poor families did not vaccinate their children**
FN4	migrant	pk	0.0011	...declared itself a sanctuary to **immigrants it considers persecuted** by President Trump
FN5	refugee	my	0.0011	...shining a spotlight on **marginalised communities such as refugees** and migrant workers

Discussion. The FP examples suggest the model over-associates *aid/support language* and group mentions with patronizing framing, even when the tone is neutral, inspirational, or informational. The FN examples show that institutional or policy-reporting language — even when genuinely patronizing — lacks the surface cues the model has learned to trigger on, leading to missed positives.

Slice-based evaluation by keyword and country

Keyword slices. Table 9 lists all keywords with at least 25 instances, sorted by F1. `women` and `immigrant` are the hardest keywords, both with recall below 0.30. At the other end, `in-need` reaches F1 = 0.79, driven by exceptionally high recall (0.91), likely because the keyword strongly correlates with the surface cues the model has learned.

Table 9: All keywords by dev F1 (min 25 examples), sorted ascending.

Keyword	n	Pos rate	Precision	Recall / F1
women	233	0.060	0.500	0.286 / 0.364
immigrant	218	0.032	1.000	0.286 / 0.444
disabled	194	0.072	0.667	0.429 / 0.522
migrant	206	0.024	1.000	0.400 / 0.571
vulnerable	209	0.096	0.571	0.600 / 0.585
hopeless	217	0.120	0.600	0.577 / 0.588
poor-families	190	0.200	0.611	0.579 / 0.595
refugee	188	0.069	0.615	0.615 / 0.615
homeless	212	0.137	0.645	0.690 / 0.667
in-need	226	0.146	0.698	0.909 / 0.789

Country slices. Table 10 shows the four worst-performing countries (min 25 examples). Australia (au), Bangladesh (bd), and the Philippines (ph) all tie at F1 = 0.43, suggesting

domain shift — likely stylistic differences across news sources — rather than a single systematic failure mode.

Table 10: Worst-performing countries by dev F1 (min 25 examples).

Country	n	Pos rate	Precision	Recall / F1
au	98	0.051	0.333	0.600 / 0.429
bd	103	0.068	0.429	0.429 / 0.429
ph	105	0.143	0.462	0.400 / 0.429
lk	85	0.106	0.444	0.444 / 0.444

Taxonomy-level (multi-task) error analysis

Because our model is trained with a 7-way multi-label taxonomy head, we analyse recall *per taxonomy category* among positive examples. Table 11 reports category-wise recall using the actual category names from the dataset. *The Saviour* is by far the hardest category (recall = 0.45), which is unsurprising given it has only 11 positive instances in the dev set. *Presupposition* is also difficult (0.58), likely because it operates at a discourse level rather than through overt lexical cues. This supports the motivation for multi-task learning: the taxonomy head pushes the encoder to capture distinct modes of patronization, but rare and pragmatically subtle categories remain challenging.

Table 11: Per-category recall on positive examples (lower = model struggles most).

Category	# positives	False negatives	Recall
The saviour	11	6	0.455
Presupposition	62	26	0.581
Authority voice	38	14	0.632
Compassion	106	36	0.660
Unequal treatment	142	45	0.683
Metaphor	52	16	0.692
Shallow solution	36	11	0.694

Takeaways. Three key behaviours emerge: (1) threshold calibration is essential — the optimal $\tau = 0.10$ is far below 0.5, and F1 degrades only gradually above it; (2) the model's strongest confusions are between genuine patronization and neutral "aid" language (FP), and between subtle or policy-embedded patronization and factual reporting (FN); and (3) performance is uneven across keywords and countries, with `women` and `immigrant` and several Global South country codes being systematically harder — suggesting that the model's learned PCL cues are skewed towards specific surface patterns associated with only some vulnerable groups.

Appendix: High-confidence error examples (full text)

False positives (full text)

FP1 (`in-need`, ph, $p = 0.9926$)
"I always consider this job as a gift, being a nurse is a reward and task given by God to

help those who are in need. Seeing your patient recover from an illness, watching their families smile when you give them care, and hearing the first cry of a newborn are just some of the things that make my work special. It might be a heavy work but it can lighten your heart," she expressed.

FP2 (`hopeless`, ph, $p = 0.9899$)
But more than taking personal strength from the heroines of Philippine culture, Alma finds greater inspiration in being able to share her skills to build on the dreams of women and children. "I look at art not as a career but as a spiritual expression. Art should bring out what is innately beautiful, especially to those who are hopeless."

FP3 (`in-need`, my, $p = 0.9893$)
Jesus begins his teaching in Matthew with the Sermon on the Mount. One group he blesses is those in need of comfort. Blessed are they who mourn, for they will be comforted (Mt 5:4).

FP4 (`in-need`, in, $p = 0.9888$)
Christmas is celebration of the birth of not merely a child, but a child who changed the destiny of humans forever. It is celebration of the fact that God wanted to be a part of the human race and so he took on flesh and blood and became human like us. We can also show unconditional love through our good deeds and helping those who are in need of our help and care. Be human and merciful.

FP5 (`refugee`, ke, $p = 0.9869$)
E-mail Address: * <h> A clinic called "Hope" helps a Syrian refugee boy cope with diabetes.

False negatives (full text)

FN1 (`vulnerable`, ng, $p = 0.0010$)
He said: "I think we can consider introducing a negligible telecom surcharge to be entirely borne by the initiator of a call. In order to protect the poor and vulnerable amongst us, we could structure it to only take effect after the third minute of talk."

FN2 (`disabled`, hk, $p = 0.0010$)
Cheung said 20 disabled undergraduate students from seven universities will start their eight-week internship in government departments this month, receiving the same salaries as able-bodied colleagues of HK$9,600 a month.

FN3 (`poor-families`, us, $p = 0.0011$)
The World Health Organization did not give a reason for the increase in deaths, but a provincial health official in Sindh said that the disease hit areas where poor families did not vaccinate their children.

FN4 (`migrant`, pk, $p = 0.0011$)
Famously progressive San Francisco has among America's most eco-friendly public policies and has declared itself a sanctuary to immigrants it considers persecuted by President Donald Trump's administration.

FN5 (`refugee`, my, $p = 0.0011$)
IdeaRaya spokesman Maryam Ramli Lee in an interview with Malay Mail Online, September

10, 2015. For human rights, Idearaya is shining a spotlight on marginalised communities such as refugees and migrant workers, while its focus for philosophy this year would be on educating children to express themselves philosophically.